EPSR-test-ex28 makemole: building the EPSR-test molecules

Program(s): makemole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `makemole`, which converts a molecule template (`.mol`, here of the `.gmol` type) into a single-molecule `.ato` file: it turns the bond lengths and bond angles into the intramolecular distance constraints that EPSR uses to keep molecules flexible but intact, sets up rotational groups and attaches the reference-potential parameters [1]. These single molecules are the building blocks from which `makeato`/`mixato` build the simulation box.

Code exercised

Fortran (original, fortran_src/):

- `makemole.f` -> `ato_manipulation_rev4.f`: `makemole` (template reading, constraints, `changelabel`, `potential` lines, box from `density`)
- `ato_manipulation_rev4.f`: `get_rotgrp` (atoms that move with a rotation about a bond), `ftype`
- `ato_read_and_save.f`: `moloutc` (writing the `.ato` file)

Rust (port, rust/epsr-core/src/):

- `tools/makemole.rs`: `run`, `get_rotgrp`, `get_whole_rotgrp`
- `ato.rs`: `ftype`, `moloutc`

Physical system

The two molecules of EPSR-test. Dimethyl sulphoxide: 10 atoms (S, two C, O, six H) with 9 bonds, 15 angles, the two methyl rotational groups (`rot 1 2`, `rot 1 3`) and Lennard-Jones + charge parameters (S: epsilon 0.84517 kJ/mol, sigma 3.56 Å, q = +0.54 e; O: q = -0.54 e). Water: O1 and two H1 with O-H 0.976 Å and H-O-H 104.5°, SPC/E Lennard-Jones and charges (epsilon 0.65 kJ/mol, sigma 3.166 Å, q(O) = -0.8476 e) [2]. Both files specify 300 K, `vibtemp 65`, `density 0.001` atoms/Å^3, so the single molecules sit in cubic boxes of 21.544 Å (DMSO) and 14.422 Å (water).

Inputs

`DMSO.mol` and `WaterMolecule.mol` copied from EPSR-test; the second command gives the file name without the `.mol` extension. No standard input is needed because every bond, angle and potential is specified.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass of two `makemole` calls; compared files: `DMSO.ato` and `WaterMolecule.ato` (the rewritten `.mol` files come out unchanged). The `.ato` files contain the per-atom constraint lists (e.g. 9 constraints for S) and the rotational groups, so identical files verify the constraint derivation. They differ from the `.ato` files stored in EPSR-test, which were processed further after `makemole`.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).

[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.